SET

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Equilibrium uniqueness and representation

In nonlinear models, standard sufficient conditions for uniqueness can be difficult to verify.

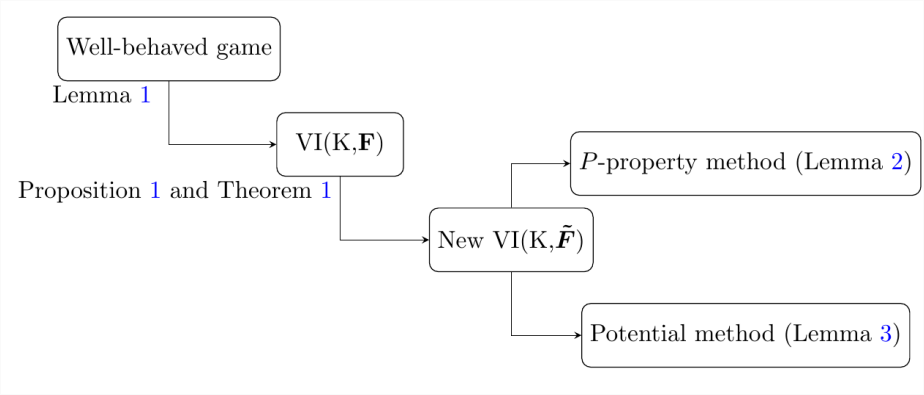
Some of this difficulty may come from how the equilibrium conditions are written.

The central question

How can we transform nonlinear equilibrium to simplify analysis, without changing the equilibrium set?

The paper develops sign-equivalent transformations (SET) to reveal useful mathematical structure in equilibrium problems.

The paper's map



An equilibrium is a best-response fixed point. The figure follows its VI representation through a sign-equivalent transformation to two methods for studying uniqueness.

Example 1: a two-player game

Two players choose $x_i \geq 0$, so $K = \mathbb{R}^2_+$.

Payoffs

$$u_1(x) = x_1\left(1 + \tfrac{1}{2}x_2\right) - \tfrac{1}{2}x_1^2,$$
$$u_2(x) = 4\left[x_2\left(1 + \tfrac{1}{2}x_1\right) - \tfrac{1}{2}x_2^2\right].$$

Each payoff is strictly concave in the player's own action.

Best responses

$$B_1(x_2) = 1 + \tfrac{1}{2}x_2, \qquad B_2(x_1) = 1 + \tfrac{1}{2}x_1.$$

The factor 4 changes player 2's payoff units, but leaves the best response unchanged.

Example 1: the VI and its transformation

For a candidate $x^* \in K$, use the negative marginal payoffs:

$$F_i = -\partial_i u_i, \qquad F(x) = \begin{pmatrix} x_1 - 1 - \frac{1}{2}x_2 \\ 4(x_2 - 1 - \frac{1}{2}x_1) \end{pmatrix}.$$

$$x^* \in \mathrm{NE} \iff \langle y - x^*, F(x^*) \rangle \geq 0 \quad \text{for every } y \in K.$$

SET: divide the second component by 4

$$\widetilde{F}(x) = \mathrm{diag}(1, \tfrac{1}{4})F(x) = \begin{pmatrix} x_1 - 1 - \frac{1}{2}x_2 \\ x_2 - 1 - \frac{1}{2}x_1 \end{pmatrix}.$$

Positive scaling preserves every component's sign on K. Since K is rectangular,

$$\mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).$$

Example 1: the two uniqueness routes

The transformed system has a symmetric derivative:

$$H = D\widetilde{F} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}, \qquad \lambda(H) \in \{\tfrac{1}{2}, \tfrac{3}{2}\}.$$

P-property method

The principal minors of H are $1, 1, \frac{3}{4} > 0$.

The P-matrix condition gives at most one VI solution.

Potential method

$$\Phi(x) = \tfrac{1}{2}(x_1^2 + x_2^2) - \tfrac{1}{2}x_1x_2 - x_1 - x_2.$$

$\nabla\Phi = \widetilde{F}$ and $\nabla^2\Phi = H \succ 0$. The VI minimizes the strictly convex Φ over K.

$$\widetilde{F}(x^*) = 0 \quad \Longrightarrow \quad \boxed{x_1^* = x_2^* = 2}.$$

This feasible zero proves existence. The equilibrium is unique.

Example 2: different nonlinear best responses

Two players choose $x_i \geq 0$, with $K = \mathbb{R}^2_+$.

Payoffs

$$u_1(x) = x_1(1 + x_2) - x_1^2 - \tfrac{1}{3}x_1^3,$$
$$u_2(x) = x_2 \ln(2 + x_1) - \tfrac{1}{2}x_2^2.$$

Each payoff is strictly concave in the player's own action.

Best responses

$$B_1(x_2) = \sqrt{2 + x_2} - 1, \qquad B_2(x_1) = \ln(2 + x_1).$$

Every equilibrium is interior because both best responses are positive on K.

Example 2: a nonlinear sign-equivalent transformation

The negative marginal payoffs define $F_i = -\partial_i u_i$.

Original VI

$$F(x) = \begin{pmatrix} x_1^2 + 2x_1 - 1 - x_2 \\ x_2 - \ln(2 + x_1) \end{pmatrix}.$$

$$\partial_2 F_1 = -1, \quad \partial_1 F_2 = -\frac{1}{2 + x_1}.$$

After SET

$$\widetilde{F}(x) = \begin{pmatrix} x_1^2 + 2x_1 - 1 - x_2 \\ e^{x_2} - 2 - x_1 \end{pmatrix}.$$

$$\partial_2 \widetilde{F}_1 = \partial_1 \widetilde{F}_2 = -1.$$

The exponential is strictly increasing:

$$x_2 \gtreqless \ln(2 + x_1) \quad \Longleftrightarrow \quad e^{x_2} \gtreqless 2 + x_1.$$

All component signs agree on the rectangular domain K. Thus

$$\mathrm{NE} = \mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).$$

Example 2: a convex equilibrium problem

The transformed operator is a gradient, $\widetilde{F} = \nabla\Phi$, where

$$\Phi(x) = \tfrac{1}{3}x_1^3 + x_1^2 - x_1 - x_1x_2 + e^{x_2} - 2x_2.$$

For every $x \in K$,

$$\nabla^2\Phi(x) = \begin{pmatrix} 2 + 2x_1 & -1 \\ -1 & e^{x_2} \end{pmatrix} \succeq \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \succ 0.$$

Unique equilibrium

Strong convexity and coercivity give a unique minimizer on K.

SET exposes a convex optimization representation of the same equilibrium problem.

Individual optimality and the VI

Definition 1 (Well-behaved game)

A normal-form game $\Gamma = (u_i, K_i)_{i \in N}$ is well-behaved if, for every player i:

1. $K_i \subseteq \mathbb{R}^{m_i}$ is closed and convex.
2. u_i is twice continuously differentiable (C^2).
3. $u_i(\cdot, x_{-i})$ is concave on K_i for every feasible x_{-i}.

Lemma 1 (Nash equilibrium and the VI)

For a well-behaved game, set $K = \prod_i K_i$ and $F_i = -\nabla_{x_i} u_i$. Then, for $x^* \in K$,

$$x^* \in \mathrm{NE} \quad \Longleftrightarrow \quad \langle y - x^*, F(x^*) \rangle \geq 0 \quad \text{for every } y \in K.$$

Sign equivalence on rectangular domains

Definition 2 (Sign equivalence)

F and $\widetilde{F}$ are sign-equivalent on K if $\operatorname{sgn} F_i(x) = \operatorname{sgn} \widetilde{F}_i(x)$ for every i and every $x \in K$.

Theorem 1 (Part (ii): rectangular domains)

If $K = \prod_i [a_i, b_i]$ and $F, \widetilde{F}$ are sign-equivalent on K, then

$$\operatorname{Sol}(K, F) = \operatorname{Sol}(K, \widetilde{F}).$$

Two constructions of a sign-equivalent operator

Proposition 1 (SET 1 and SET 2)

Finite compositions of the following componentwise operations preserve signs.

1. **Positive rescaling:**

$$\widetilde{F}_i(x) = \alpha_i(x) F_i(x), \qquad \alpha_i(x) > 0 \quad \text{on all of } K.$$

2. **An increasing transformation:** if $F_i = A_i - B_i$ and h_i is strictly increasing,

$$\widetilde{F}_i = h_i(A_i) - h_i(B_i) \quad \Longrightarrow \quad \operatorname{sgn} \widetilde{F}_i = \operatorname{sgn} F_i.$$

For strictly decreasing h_i, reverse the order: $h_i(B_i) - h_i(A_i)$.

The rescaling factor may depend on the full profile, as in oligopoly pricing. Our Example 2 uses $h(t) = e^t$. Logs and inverses require valid domains.

The role of rectangularity: a counterexample

Consider a VI with a shared constraint rather than separate action intervals:

$$K = \{x \geq 0 : x_1 + x_2 \leq 1\}, \qquad F = (-1, -1), \qquad \widetilde{F} = (-2, -1).$$

Both operators have the same component signs at every feasible point. For a constant operator, the VI minimizes its linear objective over K.

$$\mathrm{Sol}(K, F) = \{x \geq 0 : x_1 + x_2 = 1\}, \qquad \mathrm{Sol}(K, \widetilde{F}) = \{(1, 0)\}.$$

Discussion

Why does changing the relative weight of the two coordinates matter when they compete for a shared resource?

This is a coupled-constraint VI illustration. Its domain is not a product of intervals.

The P-condition and constrained uniqueness

A useful global P-condition is: for every $x \neq y$ in K,

$$\max_i (x_i - y_i)\big[F_i(x) - F_i(y)\big] > 0.$$

Suppose x and y both solve the VI on a rectangle. Feasible one-coordinate deviations give, for every i,

$$(x_i - y_i)F_i(x) \leq 0, \qquad (x_i - y_i)F_i(y) \geq 0.$$

Subtracting implies

$$(x_i - y_i)\big[F_i(x) - F_i(y)\big] \leq 0 \quad \text{for every } i.$$

This contradicts the P-condition unless $x = y$. The argument covers constrained solutions as well as interior zeros.

What a better representation can reveal

Let K be convex and $\widetilde{F}$ continuously differentiable.

Lemma 2 (Part (ii): P-property)

On a rectangle K, if $D\widetilde{F}(x)$ is a P-matrix for every $x \in K$, then the VI has at most one solution.

A P-matrix has positive principal minors.

Used for: oligopoly pricing, peer effects, CES complements.

Lemma 3 (Potential)

If $D\widetilde{F}$ is symmetric and positive semidefinite on K, the VI minimizes a convex potential with $\nabla\Phi = \widetilde{F}$. If it is positive definite, there is at most one solution.

Used for: network games, public goods, multiplex interactions.

Existence needs its own argument in each application.

Selected Applications

1. **Oligopoly pricing:** derive the markup condition and trace how the demand assumptions deliver uniqueness.
2. **Network complements:** separate the existence threshold from uniqueness and relate equilibrium to centrality.
3. **Network public goods:** construct the potential and explain how convex costs change equilibrium behavior.

For each model: what does SET preserve, and which additional argument delivers the economic result?

Oligopoly: what price should each firm charge?

There are $n \geq 2$ firms, each selling one differentiated product.

- Firm i has constant unit cost c_i and chooses $p_i \in [a_i, b_i]$, with $c_i \leq a_i < b_i$.
- Demand $Q_i(p) > 0$ depends on its own price and its rivals' prices.
- Consumers can also choose an outside option, whose price p_0 is fixed.

The firm's problem

$$\max_{p_i \in [a_i, b_i]} \pi_i(p) = (p_i - c_i) Q_i(p).$$

Can firms' pricing decisions be mutually consistent in more than one way, including when a price floor or cap binds?

Where demand comes from

Each consumer chooses the option with the highest utility:

$$U_i = v_i - p_i + \varepsilon_i, \qquad i = 0, 1, \ldots, n.$$

Here v_i measures product appeal; ε_i captures individual tastes. Option 0 means not buying from any firm. Normalize the mass of consumers to one.

A familiar example: multinomial logit

With independent standard extreme-value taste shocks,

$$Q_i(p) = \frac{e^{v_i - p_i}}{e^{v_0 - p_0} + \sum_{j=1}^{n} e^{v_j - p_j}}.$$

A higher own price loses customers to rivals or the outside option. The paper's argument covers more than this logit example.

An optimal price balances two effects

Differentiate profit, holding rivals' prices fixed:

$$\frac{\partial \pi_i}{\partial p_i} = \underbrace{Q_i}_{\text{more revenue per customer}} + \underbrace{(p_i - c_i)\frac{\partial Q_i}{\partial p_i}}_{\text{lost customers}}, \qquad \frac{\partial Q_i}{\partial p_i} < 0.$$

At an interior optimum

$$\underbrace{p_i - c_i}_{\text{actual markup}} = \underbrace{\frac{Q_i}{-\partial Q_i / \partial p_i}}_{\text{demand-implied markup } \mu_i(p)}.$$

μ_i is the inverse own-price semi-elasticity, $1/[-\partial_{p_i} \log Q_i]$. Less price-sensitive demand supports a larger markup; rivals' prices affect this sensitivity.

SET: rewrite pricing as a markup gap

The original VI uses negative marginal profit, $F_i = -\partial \pi_i / \partial p_i$. Divide by the positive number $-\partial Q_i / \partial p_i$:

A sign-equivalent pricing condition

$$\widetilde{F}_i(p) = \frac{F_i(p)}{-\partial Q_i / \partial p_i} = \underbrace{p_i - c_i}_{\text{actual markup}} - \underbrace{\mu_i(p)}_{\text{desired markup}}.$$

- $\widetilde{F}_i < 0$: the firm wants to raise its price.
- $\widetilde{F}_i > 0$: the firm wants to cut its price.
- $\widetilde{F}_i = 0$: the two pricing effects balance.

This is a new representation of the marginal condition, not a new profit function.

Price floors and caps: signs, not just zeros

Chosen price	Marginal profit	Markup gap
Floor: $p_i = a_i$	$\partial\pi_i/\partial p_i \leq 0$	$\widetilde{F}_i \geq 0$
Interior	$\partial\pi_i/\partial p_i = 0$	$\widetilde{F}_i = 0$
Cap: $p_i = b_i$	$\partial\pi_i/\partial p_i \geq 0$	$\widetilde{F}_i \leq 0$

Under E1 (own-price log-concavity), these conditions characterize best responses. Positive rescaling preserves all three cases. On $K = \prod_i [a_i, b_i]$,

$$\mathrm{NE} = \mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).$$

The entire equilibrium set is preserved. Looking only for zeros would miss equilibria with binding bounds.

The demand restrictions behind uniqueness

Write $\ell_i = \log Q_i$. Demand is positive, C^2, and decreasing in own price.

Assumption E1 (Own-price log-concavity)

$$\frac{\partial^2 \ell_i}{\partial p_i^2} \leq 0.$$

Assumption E2 (Cross-price log-supermodularity)

$$\frac{\partial^2 \ell_i}{\partial p_i \partial p_j} \geq 0 \quad (j \neq i, \text{ including the outside option } j = 0).$$

Assumption E3 (Common-price-shift invariance)

$$Q_i(p_1 + t, \ldots, p_n + t, p_0 + t) = Q_i(p_1, \ldots, p_n, p_0), \quad i = 0, \ldots, n.$$

Why the first-order pricing conditions are sufficient

For $p_i > c_i$, profit is positive. Under E1,

$$\log \pi_i = \log(p_i - c_i) + \log Q_i.$$

$$\frac{\partial^2 \log \pi_i}{\partial p_i^2} = -\frac{1}{(p_i - c_i)^2} + \frac{\partial^2 \log Q_i}{\partial p_i^2} < 0.$$

Strict log-concavity makes the pricing incentive cross zero at most once. Thus the interior and endpoint signs characterize a best response.

The cost-price endpoint

At $p_i = c_i$, marginal profit equals $Q_i > 0$. A feasible higher price gives positive profit, so cost pricing cannot be optimal.

The argument uses log-profit concavity. Profit itself may fail to be concave.

E1 and E2: derivatives of the desired markup

Use $\ell_i = \log Q_i$ and $\mu_i = -1/\partial_{p_i}\ell_i$. Then

$$\frac{\partial \mu_i}{\partial p_j} = \frac{\partial^2 \ell_i / \partial p_i \partial p_j}{(\partial \ell_i / \partial p_i)^2}.$$

- E1: $\partial_{p_i}\mu_i \leq 0$. A higher own price weakly lowers the desired markup.
- E2: $\partial_{p_j}\mu_i \geq 0$ for $j \neq i$, including $j = 0$. A higher rival price weakly raises the desired markup.

Since $\widetilde{F}_i = p_i - c_i - \mu_i$, its Jacobian satisfies

$$J_{ii} = 1 - \partial_{p_i}\mu_i \geq 1, \qquad J_{ij} = -\partial_{p_j}\mu_i \leq 0 \quad (j \neq i).$$

E1 controls the own effect. E2 controls the cross effects.

E3: relative prices and the outside option

Common-price-shift invariance also holds for the desired markup. Differentiating with respect to the common shift gives

$$\sum_{j=0}^{n} \frac{\partial \mu_i}{\partial p_j} = 0.$$

Using the nonpositive cross effects, the firms' price Jacobian satisfies

$$\sum_{j=1}^{n} J_{ij} = 1 - \sum_{j=1}^{n} \frac{\partial \mu_i}{\partial p_j} = 1 + \frac{\partial \mu_i}{\partial p_0} \geq 1,$$

$$J_{ii} - \sum_{j \neq i} |J_{ij}| = \sum_{j=1}^{n} J_{ij} \geq 1.$$

The outside-option derivative bounds the combined strategic cross effects.

A direct uniqueness proof for the pricing VI

Take distinct $p, q \in K$, write $d = p - q$, and choose i with $|d_i| = \|d\|_\infty > 0$. The segment $q + td$ stays in the price rectangle.

$$
\begin{aligned}
d_i\big[\widetilde{F}_i(p) - \widetilde{F}_i(q)\big] &= \int_0^1 d_i \sum_j J_{ij}(q + td) d_j \, dt \\
&\geq d_i^2 \int_0^1 \left[J_{ii}(q + td) - \sum_{j \neq i} |J_{ij}(q + td)| \right] dt \\
&\geq d_i^2 > 0.
\end{aligned}
$$

This proves the global P-condition. The earlier coordinate argument therefore rules out two VI solutions, including solutions with binding price bounds. **Discussion.** Where did E1, E2, E3, and rectangularity enter this proof?

Existence and uniqueness in the pricing game

Proposition 4 (Oligopoly pricing)

Under E1–E3, the pricing game on $\prod_i [a_i, b_i]$ with $a_i \geq c_i$ has at most one pure-strategy Nash equilibrium.

The proof uses strict row diagonal dominance of the transformed markup gap.

Existence on finite price intervals

Each firm has a unique best response on its compact interval. Continuous profits make this best response continuous. The joint best-response map is a self-map of the compact price box, so Brouwer gives a fixed point.

Together, these arguments give exactly one equilibrium on the finite box. An unbounded price domain requires a separate existence argument.

Verification of the logit demand assumptions

Let Q_0 be the outside-option share and $\ell_i = \log Q_i$. For unit-slope logit,

$$\partial_{p_i} \ell_i = -(1 - Q_i), \qquad \partial^2_{p_i p_i} \ell_i = -Q_i(1 - Q_i) \leq 0.$$

For every other option, including the outside option,

$$\partial^2_{p_i p_j} \ell_i = Q_i Q_j \geq 0 \qquad (j \neq i,\ j = 0 \text{ allowed}).$$

Adding t to all prices multiplies every logit weight by e^{-t}. The common factor cancels from each share, verifying E3.

Discussion

Which parts of the uniqueness proof depend on logit's formula, and which depend only on E1–E3?

A two-firm logit example: solve for prices

Take two identical firms: $c_1 = c_2 = v_1 = v_2 = 0$ and $v_0 - p_0 = 0$.

$$Q_i(p) = \frac{e^{-p_i}}{1 + e^{-p_1} + e^{-p_2}}, \qquad -\frac{\partial Q_i}{\partial p_i} = Q_i(1 - Q_i).$$

The markup gap becomes especially simple

$$\mu_i = \frac{1}{1 - Q_i}, \qquad \widetilde{F}_i = p_i - \frac{1}{1 - Q_i}.$$

With $p_i \in [0, 4]$, look for a symmetric interior price p:

$$p = \frac{1 + 2e^{-p}}{1 + e^{-p}} \quad \Longrightarrow \quad \boxed{p_1^* = p_2^* \approx 1.227, \quad Q_1^* = Q_2^* \approx 0.185.}$$

The uniqueness result also rules out asymmetric equilibria.

The same market with price floors or caps

Keep demand and costs unchanged; change only the feasible price intervals.

Interval for both firms	Equilibrium prices	Markup gap
$[0, 4]$: no binding bound	$(1.227, 1.227)$	0
$[0, 1]$: cap at 1	$(1, 1)$	-0.269
$[1.5, 4]$: floor at 1.5	$(1.5, 1.5)$	$+0.318$

At the cap firms want higher prices; at the floor they want lower prices.

What SET buys us

Bounds change prices, but sign preservation lets the same uniqueness argument cover all three cases.

What SET changes in the uniqueness test

Keep the two-firm logit example, with zero costs and deterministic appeal.

The original derivative need not be a P-matrix

Take $p_1 = p_2 = 4$ and write $s = Q_1 = Q_2 \approx 0.017668$. For $F_i = -\partial \pi_i / \partial p_i$,

$$\frac{\partial F_i}{\partial p_i} = s(1-s)\big[2 - p_i(1-2s)\big] \approx -0.03226 < 0.$$

A negative 1×1 principal minor rules out a global P-matrix test for DF. The transformed derivative passes the test throughout the same price box.

Failure of this sufficient matrix test does not imply multiple equilibria. SET exposes a useful certificate that the original derivative does not have.

Existing application: nonlinear peer effects

A social norm need not be the average peer

$$\widetilde{x}_{-i}(\beta) = \left(\sum_j \widetilde{g}_{ij} x_j^{\beta} \right)^{1/\beta}, \qquad x_i = (1-\lambda_2) a_i + (\lambda_1 + \lambda_2) \widetilde{x}_{-i}(\beta).$$

Here $\widetilde{g}_{ij} = g_{ij}/d_i$ are row-normalized links.

Proposition 3 (Nonlinear peer effects)

For $a \gg 0$, $0 < \lambda_1 + \lambda_2 < 1$, and $0 < \lambda_1 < 1$, the system has at most one positive solution.

$\beta = 1$ recovers linear-in-means; other values change relative peer influence. Taking logs reveals the *P*-property. Existence remains a separate question.

Existing applications: network games and multiplexity

Network public goods with income

An inverse Engel curve turns heterogeneous marginal conditions into a symmetric system on an undirected graph. A potential and a spectral condition can then give uniqueness.

Several network layers

An agent chooses effort across multiple kinds of ties. A transformed system can have a potential even with cross-layer costs. Extra curvature conditions are needed for uniqueness.

The same method can reveal different mathematical structures across models.

CES peer effects on an unnormalized network

$$u_i(x) = a_i x_i - \frac{1}{2}x_i^2 + \delta x_i \Big(\sum\nolimits_j g_{ij} x_j^{\beta}\Big)^{1/\beta}.$$

Best response

$$T_i(x) = a_i + \delta S_i(x), \qquad S_i(x) = \Big(\sum\nolimits_j g_{ij} x_j^{\beta}\Big)^{1/\beta}.$$

- $a_i > 0$, $\delta > 0$, $\beta > 0$, and $x_i \geq 0$.
- G is an undirected adjacency matrix, without self-loops.
- We use a connected, nonempty network throughout this CES discussion.

What the CES parameter changes

For positive neighbor actions,

$$\frac{\partial S_i}{\partial x_j} = g_{ij} x_j^{\beta-1} S_i(x)^{1-\beta}.$$

For two linked neighbors j and k,

$$\frac{\partial S_i / \partial x_j}{\partial S_i / \partial x_k} = \left(\frac{x_j}{x_k}\right)^{\beta-1}.$$

Relative peer influence

$\beta > 1$ favors high-action peers at the margin. $0 < \beta < 1$ places relatively more marginal weight on low-action peers.

Changing β also changes the scale of this unnormalized aggregate.

CES complements: the logarithmic system

Every equilibrium satisfies $x_i = T_i(x) \geq a_i > 0$. The change of variables $z_i = \log x_i$ therefore covers the entire equilibrium set.

$$R_i(x) = x_i - T_i(x), \qquad H_i(z) = z_i - \log T_i(e^z).$$

SET on positive quantities

Because the logarithm is strictly increasing,

$$\operatorname{sign} R_i(e^z) = \operatorname{sign}[\log e^{z_i} - \log T_i(e^z)] = \operatorname{sign} H_i(z).$$

Thus $R(x) = 0$ and $H(\log x) = 0$ describe the same equilibria.

The next step uses the homogeneity of S_i to study the derivative of H.

CES uniqueness: the derivative and the proof

Write $L_i(z) = \log T_i(e^z)$, $W = DL$, and $DH = I - W$. Direct differentiation gives

$$W_{ij}(z) = \frac{\delta S_i(e^z)}{a_i + \delta S_i(e^z)} \frac{g_{ij} e^{\beta z_j}}{\sum_k g_{ik} e^{\beta z_k}}, \qquad \sum_j W_{ij}(z) < 1.$$

Two distinct finite roots are impossible

If $H(z) = H(w) = 0$, then

$$z - w = \overline{W}(z - w), \qquad \overline{W} = \int_0^1 W(w + t(z - w))\, dt.$$

On this compact segment, the row sums have a common bound $\alpha < 1$. Hence $\|z - w\|_\infty \le \alpha \|z - w\|_\infty$, so $z = w$.

Discussion: where does $a_i > 0$ enter? The argument gives at most one equilibrium, without a uniform contraction bound on all of $\mathbb{R}^n$.

The spectral existence threshold

Theorem 4 (CES equilibrium)

Let $Ge = \lambda_{\max}(G)e$, $e \gg 0$, and $\bar{\delta} = \lambda_{\max}(G)^{-1/\beta}$.

(i) $0 < \delta < \bar{\delta}$: a unique positive equilibrium exists, monotone in a and δ.

(ii) As $\delta \uparrow \bar{\delta}$, all actions diverge and $x_i^*/x_j^* \to (e_i/e_j)^{1/\beta}$.

(iii) $\delta > \bar{\delta}$: no finite positive equilibrium exists.

The appendix proof also excludes $\delta = \bar{\delta}$.

Example 2 (Paper example: regular network)

For common degree $d \geq 1$, $a_i = a > 0$, and $\delta < \bar{\delta}$,

$$x_i^* = \frac{a}{1 - \delta d^{1/\beta}}, \qquad \bar{\delta} = d^{-1/\beta}.$$

CES existence: a Perron upper bound

Let $Ge = \lambda e$, with $e \gg 0$ and $\lambda = \lambda_{\max}(G)$. For $q_i = e_i^{1/\beta}$, homogeneity gives

$$S_i(q) = \lambda^{1/\beta} q_i, \qquad T_i(Mq) = a_i + \delta M \lambda^{1/\beta} q_i.$$

An invariant rectangle when $\delta < \bar{\delta}$

Choose

$$M \geq \max_i \frac{a_i}{q_i(1 - \delta\lambda^{1/\beta})}, \qquad \bar{x} = Mq.$$

Then $a \leq T(a)$, $T(\bar{x}) \leq \bar{x}$, and T is increasing. Starting from $x^{(0)} = a$, iteration increases and remains below $\bar{x}$.

The bounded sequence converges. Continuity gives $x^* = T(x^*)$. The preceding uniqueness argument makes this the only equilibrium.

CES non-existence: why the threshold is necessary

Suppose a finite equilibrium exists. Since $a \gg 0$ and $\beta > 0$,

$$(x - a)^{\beta} = \delta^{\beta} G x^{\beta} \quad \text{and} \quad (x - a)^{\beta} < x^{\beta}.$$

Powers and inequalities here apply coordinate by coordinate.

A Perron-weighted contradiction

Because G is symmetric, $e^{\top} G = \lambda_{\max} e^{\top}$. Thus

$$\delta^{\beta} \lambda_{\max} e^{\top} x^{\beta} = e^{\top} (x - a)^{\beta} < e^{\top} x^{\beta}, \qquad \delta < \lambda_{\max}^{-1/\beta} = \bar{\delta}.$$

No finite positive equilibrium can exist at or above $\bar{\delta}$.

Discussion: uniqueness of a system's roots does not guarantee that a root exists.

Comparative statics below the threshold

Theorem 4 (Part (i): comparative statics)

Within the region where a finite equilibrium exists,

$$a' \geq a \implies x^*(a', \delta, \beta) \geq x^*(a, \delta, \beta),$$

$$\delta' \geq \delta \implies x^*(a, \delta', \beta) \geq x^*(a, \delta, \beta).$$

- A higher individual baseline or stronger peer interaction raises equilibrium effort across the network.
- Equilibrium payoff is $u_i(x^*) = \frac{1}{2}(x_i^*)^2$: changes in effort also change welfare.

Bonacich centrality as the linear special case

When $\beta = 1$ and $0 < \delta < 1/\lambda_{\max}(G)$,

$$x = a + \delta G x.$$

The network multiplier

$$(I - \delta G)x = a, \qquad x^* = (I - \delta G)^{-1} a.$$

- $a = \mathbf{1}$: a standard Katz–Bonacich centrality vector.
- Heterogeneous a: baseline-weighted centrality.
- In the notation $(I - \delta G)x = b$, the vector b is this same baseline.

The network multiplier in eigenvector coordinates

Since G is symmetric, choose orthonormal eigenvectors $v_1, \dots, v_n$, with $Gv_k = \lambda_k v_k$, $v_1 = e$, and $\lambda_1 = \lambda_{\max}$.

$$a = \sum_k (v_k^\top a) v_k, \qquad x^* = (I - \delta G)^{-1} a = \sum_k \frac{v_k^\top a}{1 - \delta \lambda_k} v_k.$$

Each eigenvalue determines the amplification of its corresponding direction.

The leading direction near the threshold

Connectedness makes λ_1 simple, and $e^\top a > 0$. Therefore

$$(1 - \delta\lambda_1) x^* \longrightarrow (e^\top a) e, \qquad \frac{x^*}{\|x^*\|_2} \longrightarrow e.$$

Discussion: baseline heterogeneity remains in $e^\top a$. Why does it disappear after normalization?

The nonlinear Perron limit: proof

As $\delta \uparrow \bar{\delta}$, bounded actions would yield an equilibrium at the threshold by continuity. Connectedness propagates divergence through $x_i \geq \delta x_j$ whenever $g_{ij} = 1$. Thus every $a_i/x_i \to 0$.

$$y = x^{\beta}, \qquad q = \frac{y}{\|y\|_1}, \qquad \frac{(x-a)^{\beta}}{\|x^{\beta}\|_1} = \delta^{\beta} G q.$$

Every normalized subsequence has the same limit

For any convergent subsequence $q \to \widehat{q}$,

$$\widehat{q} = \bar{\delta}^{\beta} G \widehat{q}, \qquad \widehat{q} \geq 0, \qquad \|\widehat{q}\|_1 = 1.$$

Perron uniqueness gives $\widehat{q} = e/\|e\|_1$. Hence the full normalized sequence converges to that vector.

Taking βth roots yields the limiting ratios on the next slide.

Eigenvector centrality at the critical limit

For the positive unit Perron vector e,

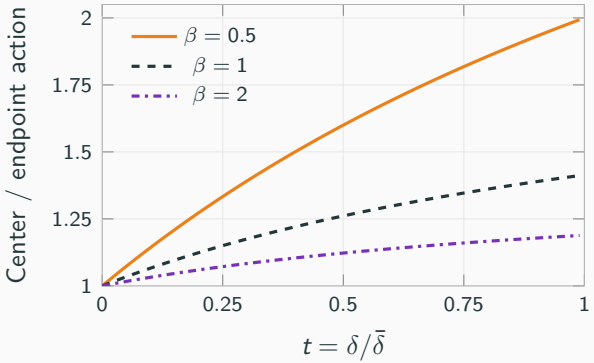
$$Ge = \lambda_{\max}(G)e, \qquad e \gg 0.$$

Theorem 4 (Part (ii): limiting ratios)

$$\lim_{\delta \uparrow \bar{\delta}} \frac{x_i^*}{x_j^*} = \left(\frac{e_i}{e_j}\right)^{1/\beta}.$$

- $\beta = 1$: $x^*/\|x^*\|_2 \to e$.
- General $\beta > 0$: the vector $(x^*)^\beta$ approaches the Perron direction.
- Baseline heterogeneity affects levels, but disappears from these limiting ratios.

Relative actions along the path



Three-node path, with common baseline $a > 0$. C is the center's action and L an endpoint's action.

$$\frac{C}{L} = \frac{1 + tq}{1 + t/q}, \qquad q = 2^{1/(2\beta)}.$$

As $t \uparrow 1$

The ratio converges to q.

The action levels diverge.

At the same fraction of the threshold, smaller β produces a larger center advantage here.

The key-player intervention

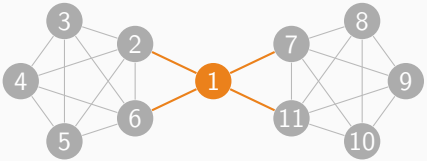
Let G^{-i} denote the network after removing player i.

Total equilibrium activity lost

$$\Delta_i = \sum_j x_j^*(G) - \sum_{j \neq i} x_j^*(G^{-i}), \qquad i^* \in \arg\max_i \Delta_i.$$

- Removing a player eliminates their action and changes everyone else's incentives.
- Highest equilibrium action need not identify the largest total effect.
- For general β, recompute equilibrium on each remaining network.

A bridge can matter more than the most active player



Linear benchmark, $\beta = 1$

δ	Key-player type
0.1	Node 2
0.2	Node 1

Nodes 2, 6, 7, and 11 have the highest degree and the highest action.

Nonlinear peers change the intervention ranking

At $\delta = 0.1$, the paper finds a switch from node 1 to the node-2 type near $\beta = 0.722$ within its reported parameter range.

Network public goods with nonlinear costs

On an undirected network, player i chooses $x_i \geq 0$ and receives

$$u_i(x) = b\left(x_i + \delta \sum_j g_{ij} x_j\right) - c(x_i), \qquad 0 < \delta \leq 1.$$

Assumption 2 (Benefits, costs, and a marginal crossing)

$$b' > 0, \qquad b'' < 0, \qquad c'(x) > 0 \ (x > 0), \qquad c'' \geq 0,$$
$$b'(0) > c'(0), \qquad \lim_{x \to \infty} b'(x) < \lim_{x \to \infty} c'(x).$$

The positive autarky effort k^* uniquely solves $b'(k^*) = c'(k^*)$.

The benchmark of Bramoullé and Kranton (2007) has $c(x) = x$ and $\delta = 1$.

Best responses, corners, and existence

Let $t_i = \delta \sum_j g_{ij} x_j$ be the public good supplied by neighbors.

If $x_i > 0$

The agent balances marginal benefit and cost:

$$b'(x_i + t_i) = c'(x_i).$$

If $x_i = 0$

The first unit is not worthwhile when

$$b'(t_i) \leq c'(0).$$

Strict concavity gives a unique continuous best response, at most k^*. The best-response mapping sends $[0, k^*]^n$ into itself. Brouwer's theorem therefore gives existence, before any spectral restriction.

The slope of the best response

Write $\theta(t) = \arg\max_{x \geq 0}\{b(x+t) - c(x)\}$. At an interior choice, differentiate $b'(\theta(t)+t) = c'(\theta(t))$:

$$b''(\theta+t)\,[\theta'(t)+1] = c''(\theta)\theta'(t), \qquad \theta'(t) = \frac{b''(\theta+t)}{c''(\theta) - b''(\theta+t)}.$$

Cost curvature. Because $b'' < 0$ and $c'' \geq 0$, $-1 \leq \theta'(t) < 0$ at an interior choice.

- Linear costs give one-for-one crowding out at interior choices.
- Positive c'' reduces the magnitude of this local response.
- At zero, further neighbor provision keeps effort at zero.

Discussion: does a slope bound for one best response prove uniqueness of the network fixed point?

The inverse marginal benefit: a valid domain

Before using $(b')^{-1}$, its argument must lie in the range of b'. Let $d_{\max}$ be the largest degree and define

$$B = \delta d_{\max} k^*, \qquad \ell = \theta(B), \qquad K = [\ell, k^*]^n.$$

For $x \in [0, k^*]^n$, $0 \le t_i \le B$, so every best response lies in K. Restricting the game to K retains its entire equilibrium set.

A finite inverse on the equilibrium-containing box

The first-order or corner condition at $\theta(B)$ gives, for $u \in [\ell, k^*]$,

$$\lim_{s \to \infty} b'(s) < c'(u) \le c'(k^*) = b'(k^*) < b'(0).$$

Thus $h(u) = (b')^{-1}(c'(u))$ is well-defined on this interval.

When $c'(0) = 0$, $\ell > 0$. No finite inverse value at zero is needed.

Substitutes: SET and a symmetric derivative

For $s_i = x_i + \delta \sum_j g_{ij} x_j$, the decreasing function b' gives

$$F_i = c'(x_i) - b'(s_i) \quad \sim_{\text{sign}} \quad \widetilde{F}_i = s_i - h(x_i), \qquad x \in K.$$

For $j \neq i$, the derivatives are

$$\frac{\partial F_i}{\partial x_j} = -\delta g_{ij} b''(s_i), \qquad \frac{\partial \widetilde{F}_i}{\partial x_j} = \delta g_{ij} = \frac{\partial \widetilde{F}_j}{\partial x_i}.$$

The original derivative is generally asymmetric. The transformed system has the minimization potential

$$\Psi(x) = \sum_i \left[\frac{x_i^2}{2} - \int_\ell^{x_i} h(u)\, du \right] + \frac{\delta}{2} x^\top G x, \qquad \nabla \Psi = \widetilde{F}.$$

SET preserves the constrained equilibria on K, including any zero-effort corners when $\ell = 0$.

Substitutes: the spectral uniqueness proof

Differentiate the inverse relation $b'(h(u)) = c'(u)$:

$$h'(u) = \frac{c''(u)}{b''(h(u))} \leq 0, \qquad D^2\Psi = I + \delta G - \mathrm{diag}(h'(x_i)).$$

For any vector v, the Rayleigh bound gives

$$v^\top D^2\Psi \, v \geq [1 + \delta\lambda_{\min}(G)]\|v\|_2^2.$$

Theorem 5 (Public-good equilibrium)

Under Assumption 2, if $1 + \delta\lambda_{\min}(G) > 0$, the paper's maximization potential ϕ^{BKC} is strictly concave and the game has a unique Nash equilibrium.

On K, $\Psi = -\phi^{\mathrm{BKC}} + \text{constant}$, using a finite integration reference. Strict convexity gives at most one solution; earlier compactness gives existence. The extra curvature $-h'$ can give uniqueness beyond the bound, as the complete-network example shows.

Zero marginal cost at zero implies full participation

Maintain Assumption 2, add strict convexity of c, and set $\delta = 1$.

Assumption 3 (Zero marginal cost at zero)

$b'(0) > c'(0) = 0$.

Proposition 7 (Part (i): full participation)

Every Nash equilibrium is interior: $x_i^* > 0$ for every player.

Proof (independent of uniqueness). At any finite profile with $x_i = 0$,

$$\left.\frac{\partial u_i}{\partial x_i}\right|_{x_i=0} = b'\Big(\sum\nolimits_j g_{ij}x_j\Big) - c'(0) > 0.$$

All equilibrium conditions therefore hold with equality:

$$b'\Big(x_i^* + \sum\nolimits_j g_{ij}x_j^*\Big) = c'(x_i^*).$$

Complete networks: convex costs remove the continuum

On a complete network with $n \geq 2$ and $\delta = 1$, write $S = \sum_j x_j$.

Proposition 7 (Part (iii))

Under Assumptions 2–3 and strictly convex costs, the equilibrium is unique:

$$x_i^* = \bar{x}, \qquad b'(n\bar{x}) = c'(\bar{x}).$$

At any equilibrium,

$$c'(x_i) = b'(S) \quad \text{for all } i.$$

Increasing c' forces equal effort; the decreasing function $b'(nz) - c'(z)$ crosses zero once.

Linear costs, $c(x) = x$

With $b'(k^*) = 1$, every profile satisfying

$$x_i \geq 0, \qquad \sum_i x_i = k^*$$

is an equilibrium.

The continuum includes both equal sharing and a single specialist.

Here $\lambda_{\min}(G) = -1$. Uniqueness with strictly convex costs holds even though $1 + \delta\lambda_{\min}(G) = 0$.

Network structure determines effort rankings

Maintain Assumptions 2–3, homogeneous b, c, strictly convex c, and $\delta = 1$. Let N_i and $\overline{N}_i = N_i \cup \{i\}$ denote open and closed neighborhoods.

Proposition 7 (Part (ii): neighborhood inclusion)

At every equilibrium,

$$N_i \subseteq N_j \quad \text{or} \quad \overline{N}_i \subseteq \overline{N}_j \quad \Longrightarrow \quad x_i^* \geq x_j^*.$$

Proper inclusion makes the effort inequality strict.

Corollary 2 (Nested split graphs)

Under the same assumptions, if G is an NSG, then $d_i < d_j$ implies $x_i^* > x_j^*$ at every Nash equilibrium.

Star, $n \geq 3$. Proposition 7(iv) gives equal leaf effort, strictly above the hub's. Degree alone does not imply this ranking on arbitrary networks.

Why the hub contributes less in a star

Maintain Assumptions 2–3, homogeneous b, c, strictly convex c, and $\delta = 1$.

Proposition 7 (Part (iv): star network)

In a star with $n \geq 3$, every equilibrium has common leaf effort $x_L > x_H$, where x_H is the hub's effort.

1. All leaves face x_H. Their unique best response gives one common $x_L > 0$.
2. The hub and a leaf satisfy
$$c'(x_H) = b'(x_H + (n-1)x_L), \qquad c'(x_L) = b'(x_H + x_L).$$
3. Since $(n-2)x_L > 0$, decreasing b' and increasing c' imply
$$c'(x_H) < c'(x_L) \quad \Longrightarrow \quad \boxed{x_H < x_L}.$$

The proof applies to every equilibrium, without a uniqueness assumption.

What SET establishes in these applications

Question	Required argument
Equilibrium equivalence	Valid VI representation, full-domain sign equivalence, and appropriate domain geometry.
Existence	A compactness, coercivity, strong-P, or bounded-iteration argument.
Uniqueness	A suitable P property, strict monotonicity, or strict convexity.

Main conclusions

A general method

On rectangular domains, the signs of the equilibrium operator determine the VI solution set. Suitable transformations can reveal structure absent from the original Jacobian.

- CES complements combine a SET uniqueness argument with a Perron existence threshold.
- Bonacich centrality is a resolvent response. Eigenvector centrality describes its critical direction.
- Under strategic substitutes, cost curvature and the smallest network eigenvalue govern uniqueness.

Which transformation exposes the useful structure in a given model?

Backup: a coordinate proof of Theorem 1(ii)

Let $K = \prod_i K_i$ and suppose $x^* \in \mathrm{Sol}(K, F)$. For every i and $t \in K_i$,

$$(t - x_i^*) F_i(x^*) \geq 0.$$

Sign equivalence implies the same inequality for $\widetilde{F}_i(x^*)$:

$$\begin{cases} F_i(x^*) = 0 & \Longrightarrow \ \widetilde{F}_i(x^*) = 0, \\ F_i(x^*) > 0 & \Longrightarrow \ t - x_i^* \geq 0 \text{ and } \widetilde{F}_i(x^*) > 0, \\ F_i(x^*) < 0 & \Longrightarrow \ t - x_i^* \leq 0 \text{ and } \widetilde{F}_i(x^*) < 0. \end{cases}$$

Consequently, for every $x \in K$,

$$\langle x - x^*, \widetilde{F}(x^*) \rangle = \sum_i (x_i - x_i^*) \widetilde{F}_i(x^*) \geq 0.$$

Thus $\mathrm{Sol}(K, F) \subseteq \mathrm{Sol}(K, \widetilde{F})$. Interchanging the two operators gives the reverse inclusion.

Backup: matrix tests in the payoff-scaling example

For $J = \begin{pmatrix} 1 & -\delta \\ -\kappa\delta & \kappa \end{pmatrix}$ with $\kappa > 0$:

P-matrix test

The principal minors are

$$1, \quad \kappa, \quad \kappa(1-\delta^2).$$

All are positive if $0 < \delta < 1$.

Strict monotonicity test

For $S = (J + J^\top)/2$,

$$\det S = \kappa - \frac{(1+\kappa)^2\delta^2}{4}.$$

$S \succ 0$ iff $\delta < 2\sqrt{\kappa}/(1+\kappa)$.

Let $\Delta = (1-\kappa)^2 + (1+\kappa)^2\delta^2$. The smallest eigenvalue is

$$\lambda_{\min}(S) = \frac{1+\kappa-\sqrt{\Delta}}{2}.$$

Rescaling by $\mathrm{diag}(1, 1/\kappa)$ restores the symmetric derivative with eigenvalues $1-\delta$ and $1+\delta$.

Backup: algebraic solutions and positive equilibria

For two linked agents with baseline $a = \mathbf{1}$,

$$G = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \lambda_1 = 1, \quad \lambda_2 = -1.$$

Whenever $\delta \neq \pm 1$, the linear system has the solution

$$x = (I - \delta G)^{-1}\mathbf{1} = \frac{1}{1-\delta}\mathbf{1}.$$

Parameter	Algebraic system	Economic equilibrium
$0 < \delta < 1$	Unique solution	Positive and finite
$\delta = 1$	Singular, inconsistent	None
$\delta > 1$	Unique negative solution	None on $\mathbb{R}^2_+$

Backup: proof of Proposition 7(iii)

Assume homogeneous b, c, strictly convex c, $c'(0) = 0$, and $\delta = 1$. On the complete network, write $S = \sum_j x_j$.

1. Full participation makes every first-order condition an equality:
$$c'(x_i) = b'(S) = c'(x_j) \qquad \text{for every } i, j.$$
2. Strictly increasing c' implies $x_i = x_j = \bar{x}$.
3. The function $f(z) = b'(nz) - c'(z)$ is strictly decreasing, with $f(0) > 0$ and an eventually negative value. Thus it has exactly one positive zero.

Why the spectral test is only sufficient

For $n \geq 2$, the complete graph has $\lambda_{\min}(G) = -1$. At $\delta = 1$ the strict spectral inequality fails, but the argument above proves uniqueness.